

BASIC COUNTING RULES – QUICK REFERENCE

1) Multiplication Rule

If something happens in steps, multiply the choices.

Example: 3 shirts and 4 pants $\rightarrow 3 \times 4 = 12$

2) Powers (Repeated Choices)

If the same number of choices repeats n times: k^n

Example: True/False for 9 questions $\rightarrow 2^9 = 512$

3) Factorial (Arranging Everything)

$n! = n \times (n-1) \times \dots \times 1$

Example: 6 people in line $\rightarrow 6! = 720$

4) Permutations (Order Matters)

$nPr = n! / (n - r)!$

Example: 8 players filling 5 positions $\rightarrow 8P5$

5) Combinations (Order Does NOT Matter)

$nCr = n! / [r! (n - r)!]$

Example: Choose 3 from 8 $\rightarrow 8C3$

6) Circular Arrangements

$(n - 1)!$

Example: 8 items in a circle $\rightarrow 7!$

7) When Items Repeat

$n! / (a! b! c! \dots)$

Example: Word with repeated letters

8) All Different Problems

Start large and count downward.

Example: $365 \times 364 \times 363 \times \dots$

HOW TO DECIDE WHAT TO USE

- Are there steps? $\rightarrow$ Multiply
- Same choice repeated? $\rightarrow$ Powers
- Arranging everything? $\rightarrow$ Factorial
- Does order matter? $\rightarrow$ Permutation
- Does order NOT matter? $\rightarrow$ Combination
- In a circle? $\rightarrow (n-1)!$
- Repeated items? $\rightarrow$ Divide by repeats
- Must all be different? $\rightarrow$ Count downward